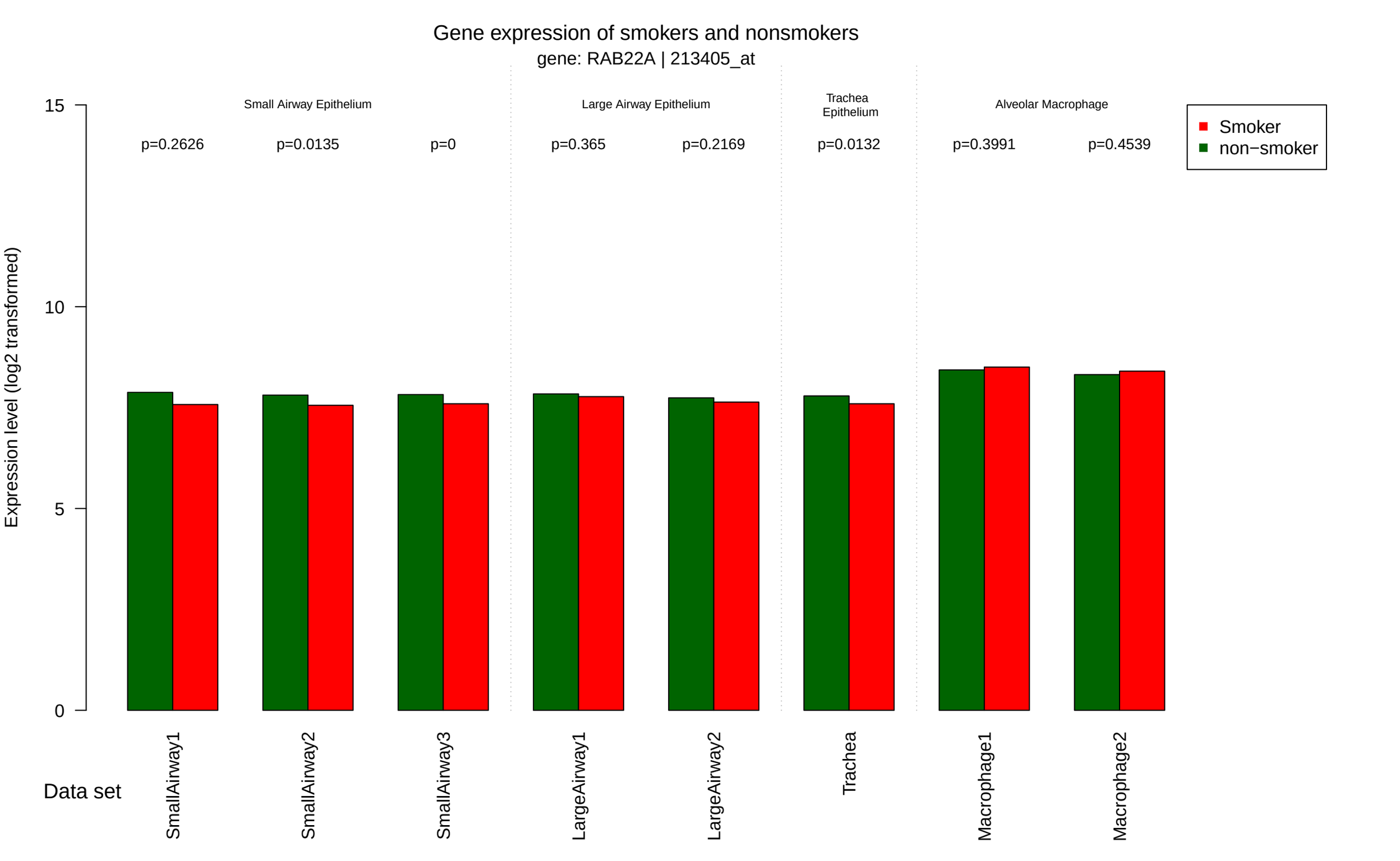
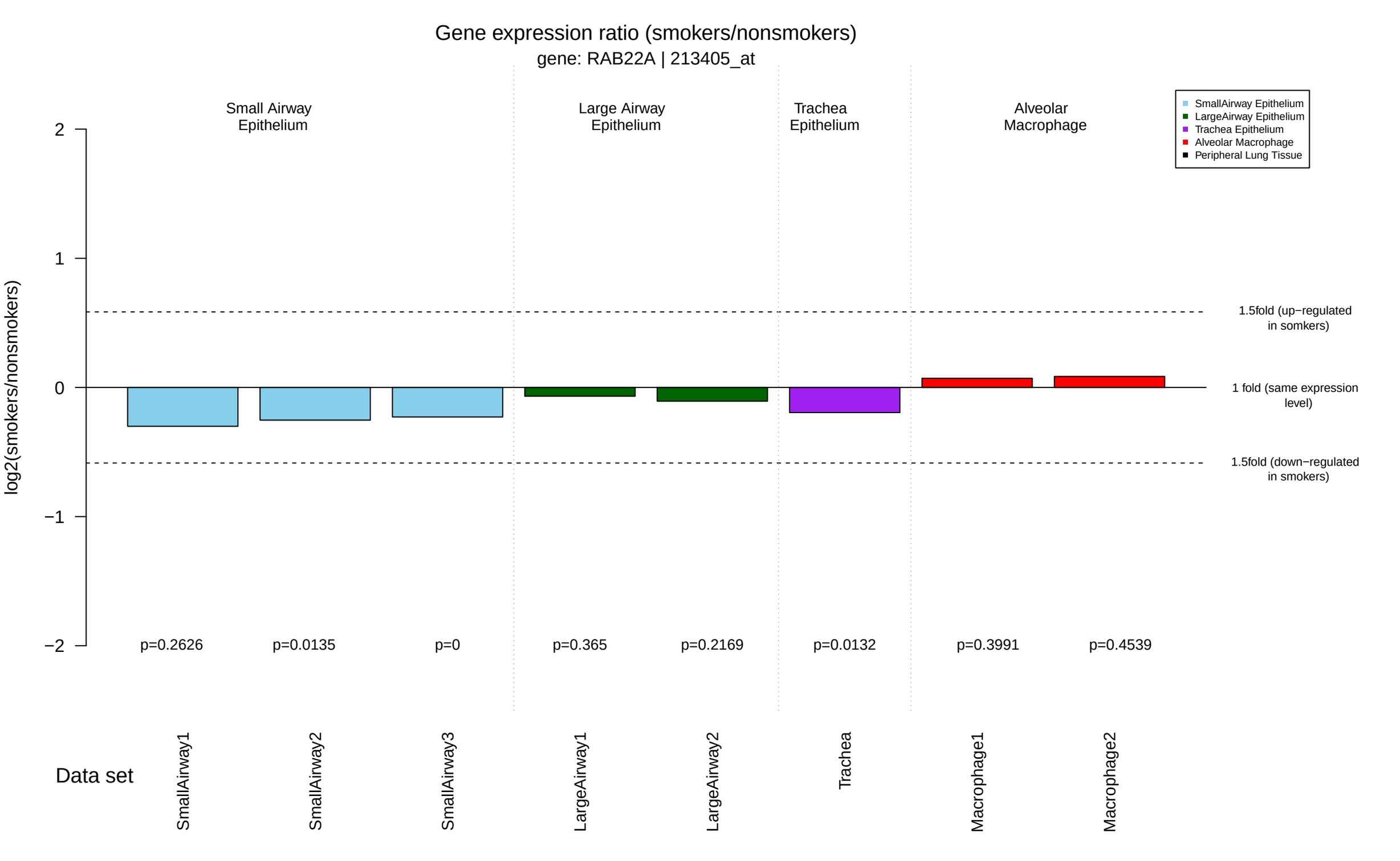
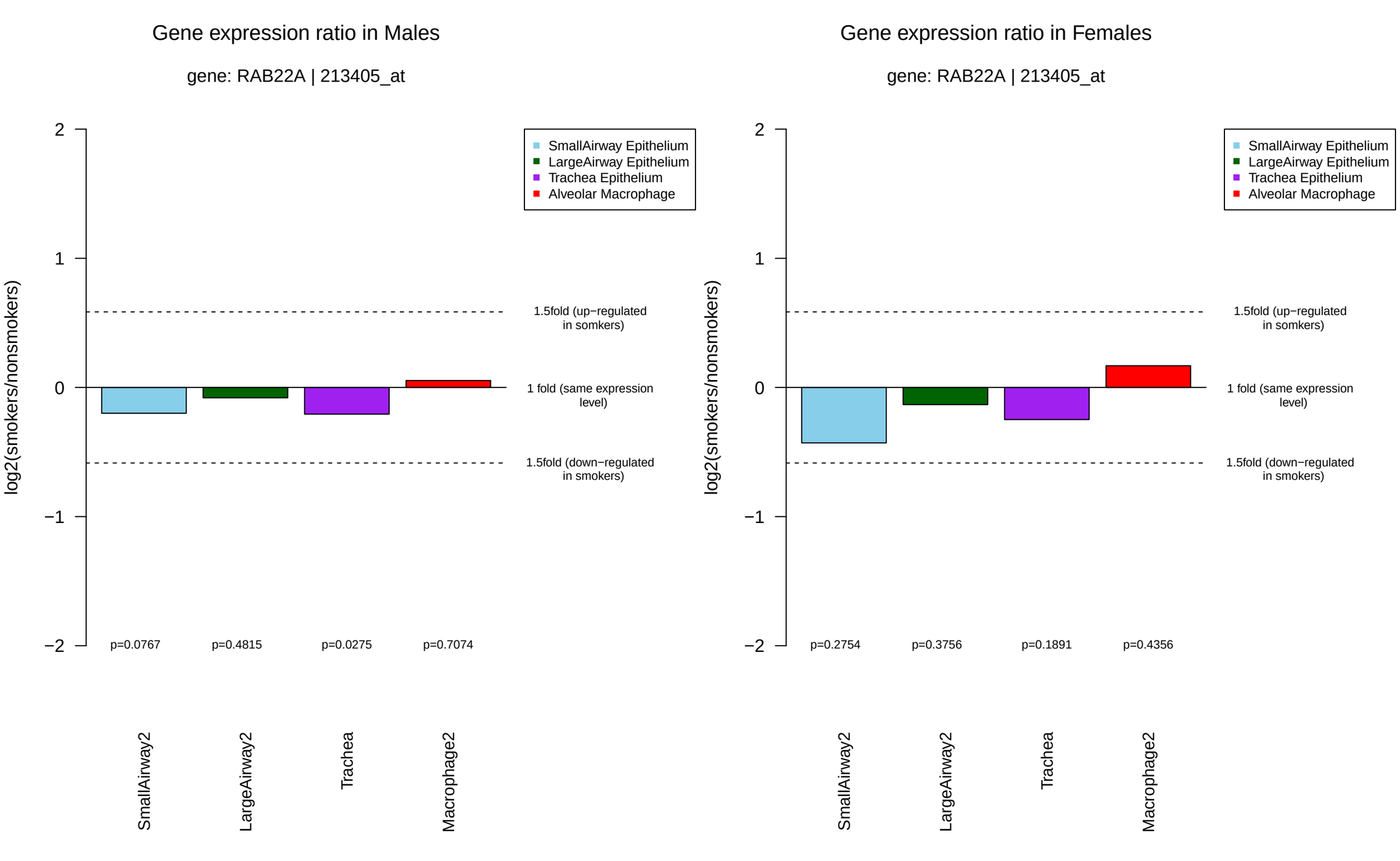
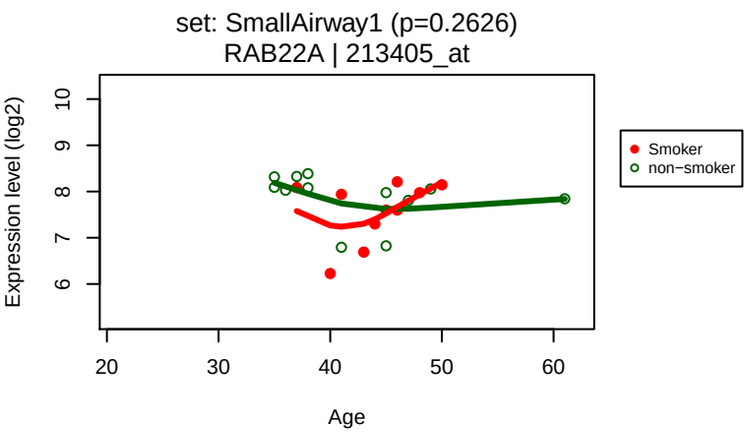
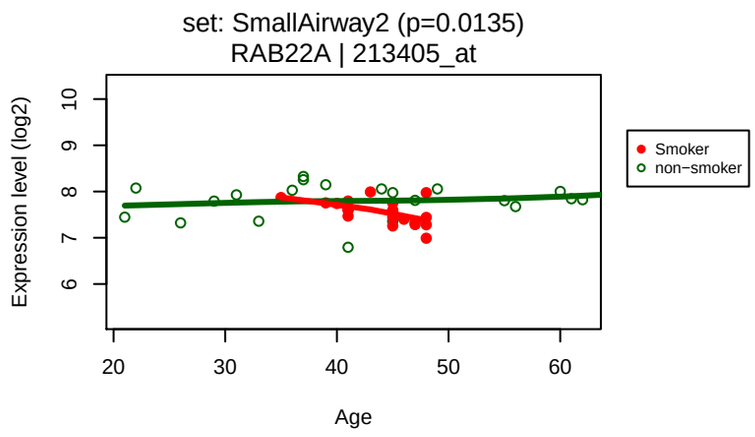
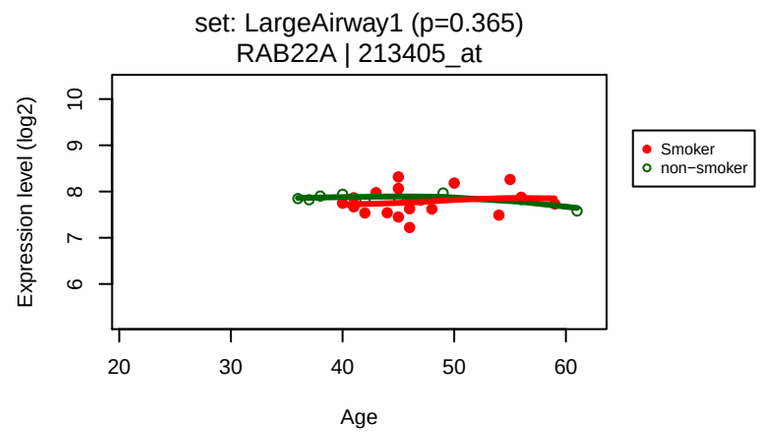

Gene expression of smokers and nonsmokers
gene: RAB22A | 213405_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2626
p=0.0135
p=0
p=0.365
p=0.2169
p=0.0132
p=0.3991
p=0.4539
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

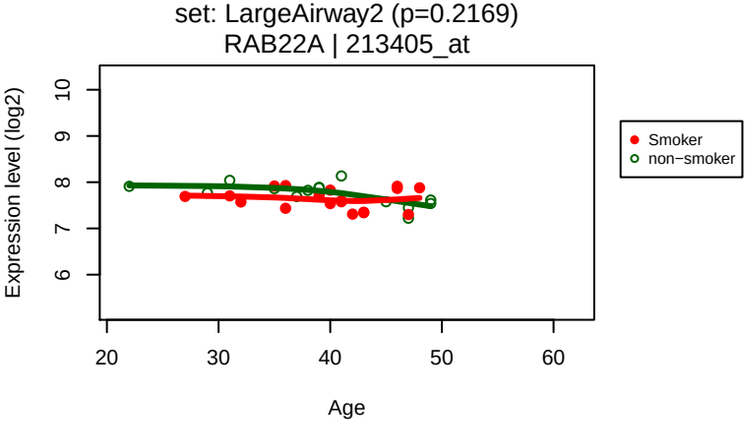

Gene expression ratio (smokers/nonsmokers)
gene: RAB22A | 213405_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2626
p=0.0135
p=0
p=0.365
p=0.2169
p=0.0132
p=0.3991
p=0.4539
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

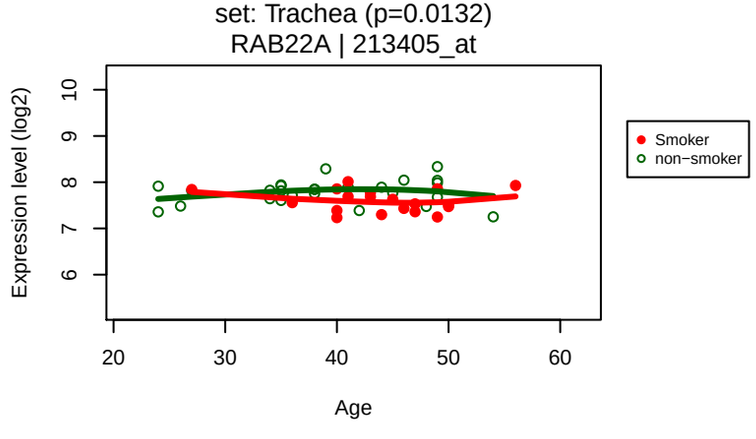

Gene expression ratio in Males
gene: RAB22A | 213405_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0767
p=0.4815
p=0.0275
p=0.7074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB22A | 213405_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2754
p=0.3756
p=0.1891
p=0.4356
SmallAirway2
LargeAirway2
Trachea
Macrophage2

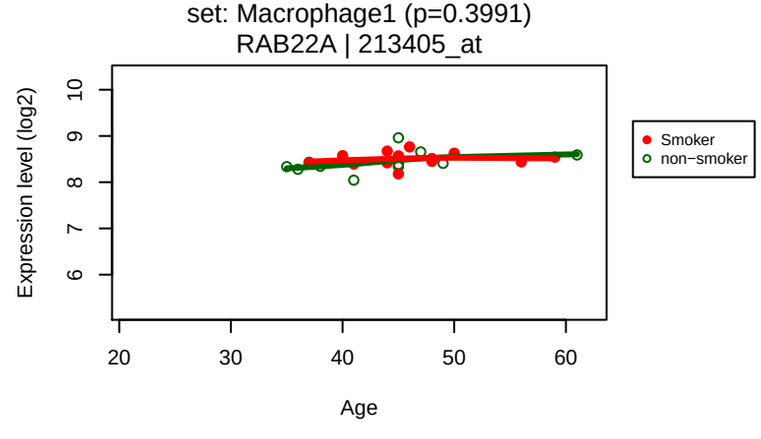

set: SmallAirway1 (p=0.2626)
RAB22A | 213405_at
Expression level (log2)
Age
Smoker
non−smoker

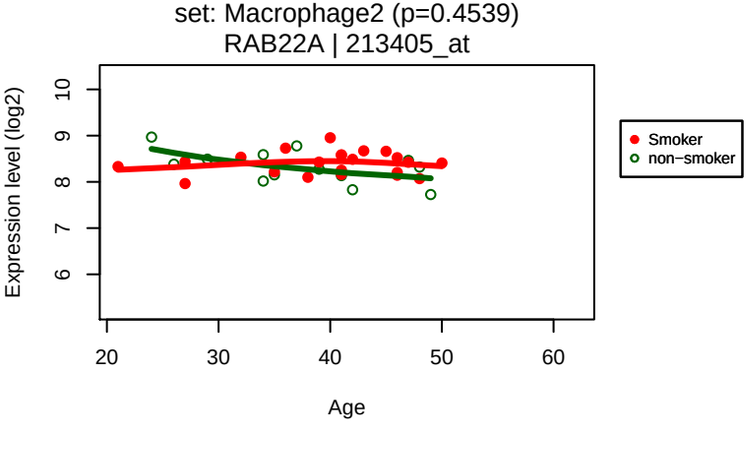

set: SmallAirway2 (p=0.0135)
RAB22A | 213405_at
Expression level (log2)
Age
Smoker
non−smoker

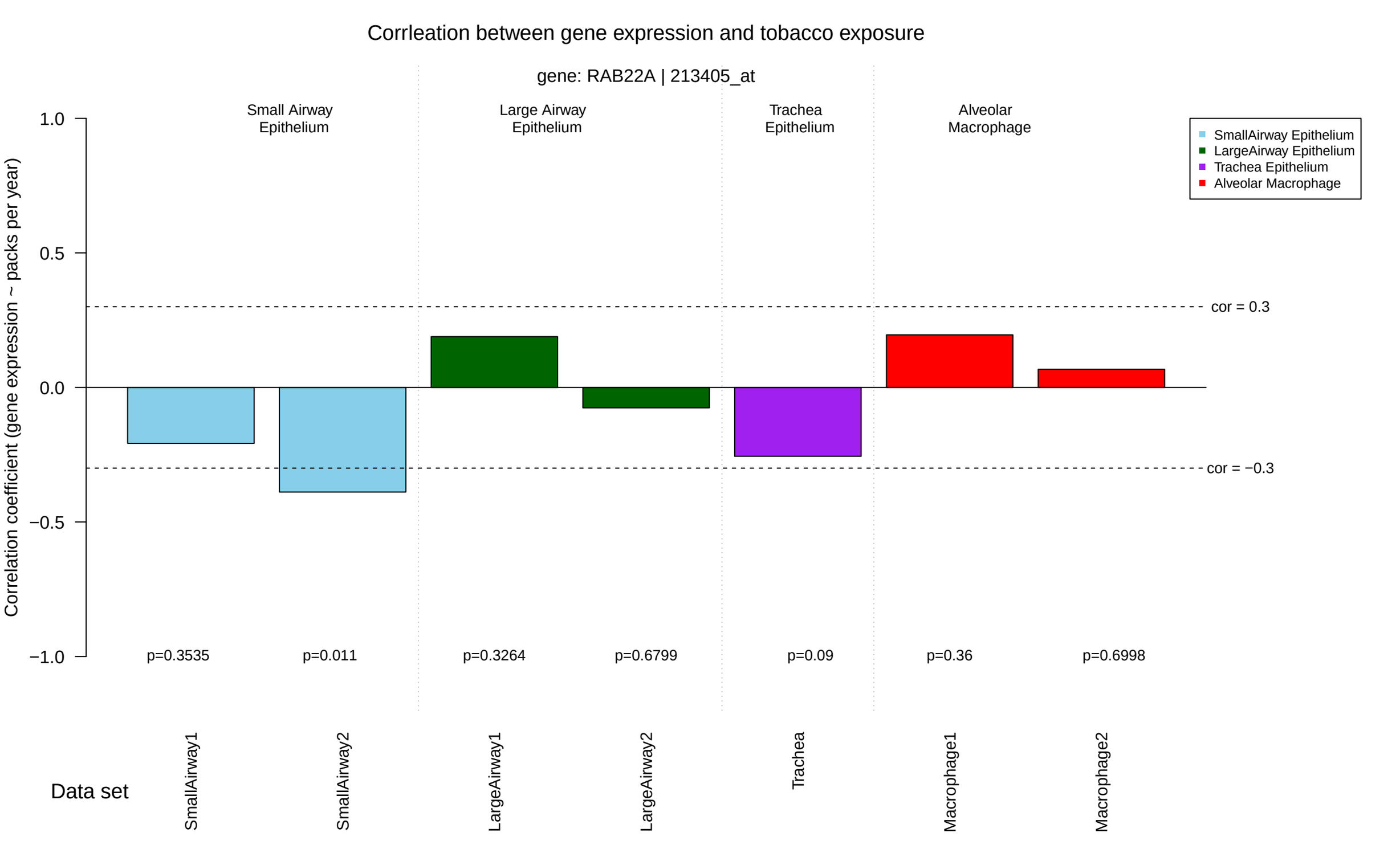

set: LargeAirway1 (p=0.365)
RAB22A | 213405_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2169)
RAB22A | 213405_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0132)
RAB22A | 213405_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3991)
RAB22A | 213405_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4539)
RAB22A | 213405_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB22A | 213405_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3535
p=0.011
p=0.3264
p=0.6799
p=0.09
p=0.36
p=0.6998
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2